

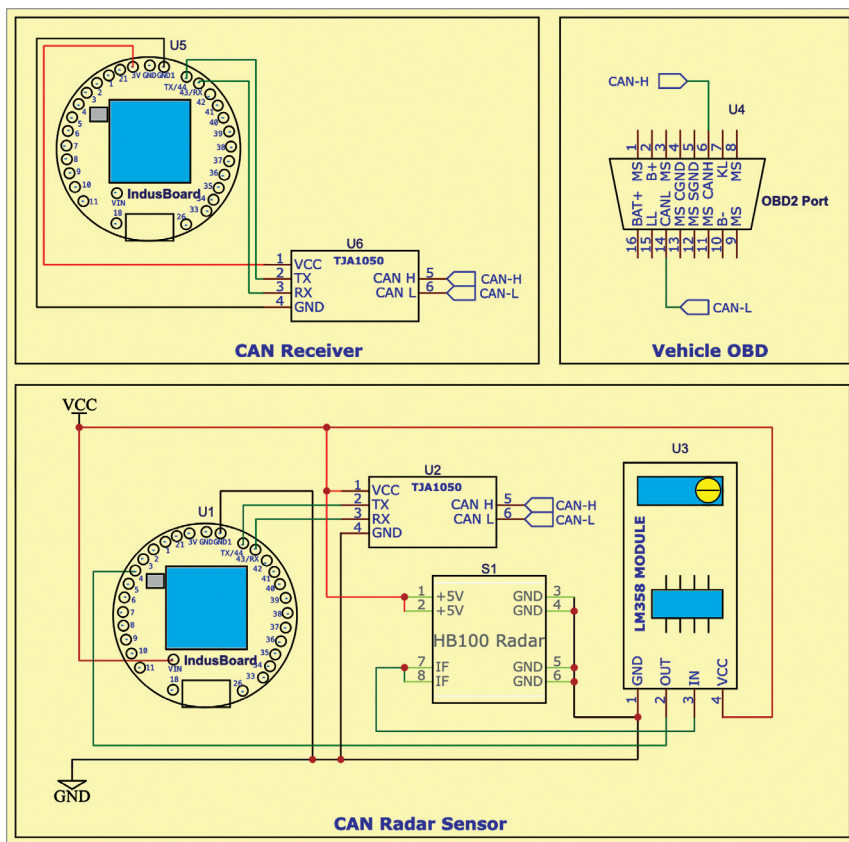


Fig. 5: CAN bus receiver, radar, and OBD port circuit

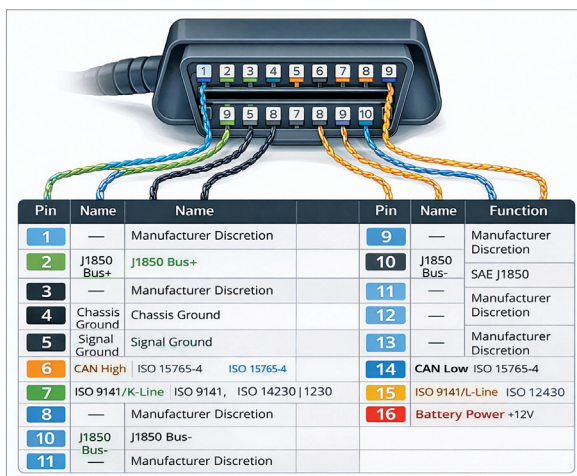


Fig. 6: OBD port pin connection

it is moving away, the frequency decreases (negative shift).

By measuring this frequency shift, the system accurately calculates the relative speed between the vehicle and the target. This information is essential for ADAS functions such as collision avoidance, adaptive cruise control, and autonomous braking. Fig. 2 shows

or another vehicle moves towards or away from it, the radar displays the speed of that approaching or receding object.

Case 2: Moving vehicle, stationary object. When the radar-equipped vehicle is moving and encounters a stationary object, such as another car, a person, or a wall, the radar shows the speed at which

it is approaching or moving away from that object.

Case 3: Two vehicles moving towards each other. When two vehicles are moving towards each other, the radar displays their relative (closing) speed, indicating how quickly they are approaching a collision.

Case 4: Two vehicles moving in the same direction. When both vehicles are moving in the same direction, the radar shows their relative speed. If both are moving at the same speed, the displayed speed is zero, indicating no relative motion. If one vehicle is faster, the radar shows the difference in their speeds, representing how quickly one is moving away from or catching up with the other.

Fig. 4 illustrates doppler radar speed measurement across different real-world scenarios, including opposite-direction movement (collision speed), same-direction movement (relative speed), and approach towards a stationary object.

Circuit and working

Fig. 5 shows the combined circuit diagram of the CAN bus radar. It is built around two IndusBoard Coin modules (U1 and U5), an LM358 amplifier (U2), an HB100 radar sensor (S1), two TJA1050 CAN bus modules (U3 and U6), and a few additional components. Most modern vehicles use the CAN protocol on these pins, with 500kbps a common speed. By connecting the IndusBoard Coin TWAI CAN bus pins along with the TJA1050 transceiver to these lines, the device can listen to or transmit messages on the vehicle's CAN network. However, caution must be exercised to avoid interference with critical vehicle systems.

In this system, the HB100 Doppler radar is utilised with the IndusBoard Coin. The Coin features a built-in IMU and temperature sensors, along with native CAN bus support via its TWAI peripheral.

The system consists of two primary units: the radar sensor